

Elements of Decision Theory: Part I

Luca Dall'Asta

DISAT
Politecnico di Torino, Italy

September 2025

Different approaches to decision making

Decision theory is a discipline at the interface of economics, philosophy, psychology and statistics, which studies the reasoning underlying single individual's choices:

- *normative decision theory*: conditions under which fully rational individuals take optimal decisions;
- *descriptive decision theory*: behavioral rules for individual choice which are consistent with experimental observations.

Our plan:

- axiomatic definition of rational decision making based on the *principle of utility maximization* (J. von Neumann and O. Morgenstern, 1944);
- generalization to sequential decisions problems according to the *principle of optimality* (R. Bellman, 1954)

Preference relations

Consider a set X of mutually exclusive ***alternatives***.

Definition

A ***preference relation*** $\succsim$ is a binary relation on the set X of alternatives such that

$$x \succsim y, \forall x, y \in X \iff x \text{ is at least as good as } y;$$

a *strict preference relation* $\succ$ is such that

$$x \succ y \iff x \succsim y \wedge \neg(y \succsim x);$$

the *indifference relation* $\sim$ is such that

$$x \sim y \iff x \succsim y \wedge y \succsim x.$$

Preference relations

Definition

The preference relation $\succsim$ is *complete* if $\forall x, y \in X, x \succsim y \vee y \succsim x$.

The preference relation $\succsim$ is *transitive* if $\forall x, y, z \in X, x \succsim y \wedge y \succsim z \implies x \succsim z$.

A preference relation that satisfies completeness and transitivity is called **rational** and it *weakly orders* the set X .

Example

$X = \{\text{icecream}, \text{sandwich}, \text{hamburger}\}$, suppose $\text{icecream} \succsim \text{sandwich}$ and $\text{hamburger} \succsim \text{icecream}$ then $\text{hamburger} \succsim \text{sandwich}$.

Remark

These concepts are still debated due to issues concerning completeness, small differences, framing problems.

Utility representation of preferences

Utility function: assigns numbers to alternatives so that the ordering of numbers reflects preferences

Definition

A function $u : X \rightarrow \mathbb{R}$ is a (*ordinal*) *utility function* representing the preference relation $\succsim$ if

$$\forall x, y \in X, \quad x \succsim y \iff u(x) \geq u(y).$$

- Ordinal utility functions represent the order of preferences, but *not the intensity*. The magnitude difference $u(x) - u(y)$ has no meaning.
- Any strictly increasing transformation (*ordinal transformation*) of u represents the same preference relation.

Utility representation of preferences

Cardinal utility functions measure also the relative preference.

Definition

A **cardinal** (or *interval-valued*) utility function is a utility function that is unique up to a positive affine transformation; i.e. if u is a cardinal utility function for $\succsim$ then $\tilde{u} = a + bu$ with $a, b \in \mathbb{R}$ and $b > 0$ is also a cardinal utility function for $\succsim$.

Remark

Cardinal utility functions are necessary for Expected Utility Theory.

Proposition

A preference relation $\succsim$ can be represented by a (ordinal/cardinal) utility function only if it is rational. Rationality is also sufficient if X is finite or countably infinite.

Decision under risk

*Decision under **risk***: the outcome of a choice is a stochastic event whose probability is **exactly known** a priori.

The set X of alternatives is replaced by:

- a set S of *states*, i.e. all possible realizations of the observable world;
- a set O of *outcomes* of the choice in different states of the world;
- a set A of *acts/actions*, i.e. functions that assign outcomes to states.

Example

$S = \{s_1, s_2\} = \{\text{meat infected, meat not infected}\},$

$A = \{a_1, a_2\} = \{\text{eat steak, eat pasta}\},$

$O = \{o_1, o_2, o_3\} = \{\text{steak \& sick, steak \& not sick, pasta}\}$

Here, $o_2 = a_1(s_2) \succ o_3 = a_2(s_1) = a_2(s_2) \succ o_1 = a_1(s_1).$

Decision under risk: lotteries

A **lottery** is a convex linear combination of outcomes induced by a probability measure.

Definition

Given a finite set of outcomes $O = \{o_i\}_{i=1,\dots,N}$ for the event under consideration, a **simple lottery** L is a list $L = (p_1, \dots, p_N)$ with $p_i \geq 0$, $\forall i$ and $\sum_{i=1}^N p_i = 1$.

It is also possible to define a lottery by randomizing over acts.

Definition

Given K simple lotteries $L_k = (p_1^k, \dots, p_N^k)$ with $k = 1, \dots, K$ and real numbers $\alpha_k \geq 0$ satisfying $\sum_{k=1}^K \alpha_k = 1$, the **compound lottery** $\{L_1, \dots, L_K; \alpha_1, \dots, \alpha_K\}$ is the risky alternative that yields the simple lottery L_k with probability α_k .

Decision under risk: lotteries

Definition

For any compound lottery $\{L_1, \dots, L_K; \alpha_1, \dots, \alpha_K\}$ it is possible to compute a corresponding **reduced lottery** $\tilde{L} = (\tilde{p}_1, \dots, \tilde{p}_N)$ with $\tilde{p}_i = \sum_{k=1}^K \alpha_k p_i^k$.

Remark

A decision maker is indifferent between a compound lottery and its reduced lottery (*consistency*). Agents only care about final probabilities, not about the structure of the lottery.

In the following, we can focus on simple lotteries $\mathcal{L}$.

Decision under risk: vN–M utility function

A lottery L induces an **expected utility function** $U(L)$ on a (finite) set of outcomes endowed with a cardinal utility function u .

Definition

The function $U : \mathcal{L} \rightarrow \mathbb{R}$ has the expected utility form (i.e. it is *additively separable*) if there is an assignment of numbers $(u_1, \dots, u_N)$ to the N possible outcomes such that, for every simple lottery $L = (p_1, \dots, p_N) \in \mathcal{L}$ we have

$$U(L) = \sum_{i=1}^N p_i u_i,$$

U is called *von Neumann–Morgenstern* (vN–M) *expected utility function*.

Decision under risk: vN–M utility function

Example

$S = \{s_1, s_2\} = \{\text{meat infected}, \text{meat not infected}\},$

$A = \{a_1, a_2\} = \{\text{eat steak}, \text{eat pasta}\},$

$O = \{o_1, o_2, o_3\} = \{\text{steak \& sick}, \text{steak \& not sick}, \text{pasta}\}$

Action a_1 corresponds to the lottery $L = (p, 1 - p, 0)$ with $\Pr[\text{steak infected}] = p$, then

$$U(L) = \mathbb{E}[u(a_1)] = pu(o_1) + (1 - p)u(o_2)$$

Any other randomization over actions and outcomes can be represented by lotteries.

Decision under risk: vN–M utility function

Proposition

A vN–M expected utility function U is **linear** in its argument, i.e. given the compound lottery $\{L_1, \dots, L_K; \alpha_1, \dots, \alpha_K\}$ we have

$$U(\{L_1, \dots, L_K; \alpha_1, \dots, \alpha_K\}) = U\left(\sum_{k=1}^K \alpha_k L_k\right) = \sum_{k=1}^K \alpha_k U(L_k).$$

Proposition

Suppose that $U : \mathcal{L} \rightarrow \mathbb{R}$ is a vN–M expected utility function for the preference relation $\succsim$ on $\mathcal{L}$, then $\tilde{U} : \mathcal{L} \rightarrow \mathbb{R}$ is another vN–M expected utility function for $\succsim$ if and only if $\exists \gamma, \beta \in \mathbb{R}$ with $\beta > 0$ such that

$$\tilde{U}(L) = \gamma + \beta U(L), \quad \forall L \in \mathcal{L}.$$

Decision under risk: vN–M utility function

Definition

The preference relation $\succsim$ on $\mathcal{L}$ is **continuous** if $\forall L_1, L_2, L_3 \in \mathcal{L}$ such that $L_3 \succsim L_2 \succsim L_1$, there exists $p \in [0, 1]$ such that the compound lottery $\{L_1, L_3; p, 1 - p\}$ satisfies the relation

$$\{L_1, L_3; p, 1 - p\} \sim L_2.$$

Definition

The preference relation $\succsim$ on $\mathcal{L}$ satisfies the **independent axiom** if $\forall L_1, L_2, L_3 \in \mathcal{L}$ and $\alpha \in (0, 1)$ we have

$$L_2 \succsim L_1 \iff \{L_2, L_3; \alpha, 1 - \alpha\} \succsim \{L_1, L_3; \alpha, 1 - \alpha\}.$$

Decision under risk: Expected Utility Theorem

von Neumann-Morgenstern Theorem

Consider a finite set of outcomes $O = \{o_i\}_{i=1}^N$ and the set of all simple lotteries $\mathcal{L}$ defined on O . Suppose that the preference relation $\succsim$ is (1) rational and satisfies (2) continuity and (3) independence, then it admits a utility representation of the (vN-M) expected utility form; that is, there exist a set of numbers $\{u_1, u_2, \dots, u_N\}$ such that for any two lotteries L, L' with $L = (p_1, \dots, p_N)$ and $L' = (p'_1, \dots, p'_N)$

$$L \succsim L' \iff \sum_{n=1}^N u_n p_n \geq \sum_{n=1}^N u_n p'_n.$$

Corollary (PRINCIPLE EXP. UTIL. MAXIM.)

A rational agent facing a decision under risk will behave as if she is maximizing the value of a corresponding vN-M expected utility.

Risk propensity

Remark (D. BERNOULLI, 1713)

A fair coin is tossed until it gives a head. If the head comes at the n -th trial, you are given a prize equal to 2^n \$.

How much would you be willing to pay to play the gamble?

Suppose that a simple lottery is represented by a c.d.f. $F : O \rightarrow [0, 1]$ on “monetary” outcomes $O = (\underline{x}, \bar{x}) \subseteq \mathbb{R}$, the expected utility of the lottery F is

$$U(F) = \int_{\underline{x}}^{\bar{x}} u(x) dF(x).$$

Risk neutrality: Utility function is **linear** in wealth, i.e. $u(x) = x$

It means that agents only care about the expected value, not about its distribution. This is not realistic (see St. Petersburg's Paradox).

Risk propensity

Restrictions on $u(x)$ are necessary: u is assumed to be a *continuous*, *increasing* and **concave** function of x .

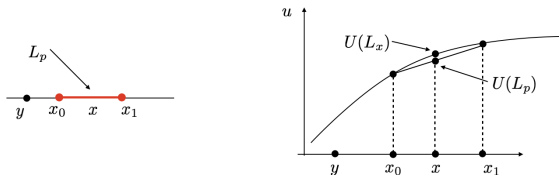
CONCAVITY: Consider the lottery $L_p = (p, 1 - p)$ with “monetary” reward $x = px_0 + (1 - p)x_1$ with $x_0 \leq x_1$ and $u(\cdot)$ such that

$$u(x) \geq pu(x_0) + (1 - p)u(x_1),$$

then it also holds

$$U(L_p) = pU(L_0) + (1 - p)U(L_1) \leq U(L_x) = u(x)$$

where L_0, L_1 are lotteries with $p = 0$ and $p = 1$, L_x puts full weight on x .



Risk propensity

Definition

A decision maker is **risk-averse** if and only if for any lottery F , the degenerate lottery F_m that yields for sure the amount $x_m = \int_{\underline{x}}^{\bar{x}} x dF(x)$ is at least as good as the lottery F , or equivalently if and only if

$$U(F) = \int_{\underline{x}}^{\bar{x}} u(x) dF(x) \leq u(x_m) = U(F_m), \quad \forall F.$$

Example

Reconsider the St. Petersburg's problem with $u(x) = \sqrt{x}$, then

$$\begin{aligned} \mathbb{E}[u(\text{play})] &= \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i u(2^i) \\ &= \sum_{i=1}^{\infty} 2^{-i/2} = 1 + \sqrt{2} < +\infty. \end{aligned}$$

Violation of independence

Remark (M. ALLAIS, 1953)

Experiments uncovered some limitations of Expected Utility Theory by common violations of the independence axiom.

Consider four lotteries (two groups of individuals)

- L_A gives 1M dollars for sure;
- L'_A gives 1M with prob. 0.89, 5M with prob. 0.1 and nothing otherwise;
- L_B gives 1M dollars with prob. 0.11 and nothing otherwise;
- L'_B gives 5M with prob. 0.1 and nothing otherwise.

	0.89	0.1	0.01
L_A	1M	1M	1M
L'_A	1M	5M	0M
L_B	0M	1M	1M
L'_B	0M	5M	0M

Which one do you prefer between L_A and L'_A ? and between L_B and L'_B ?

Violation of independence

Experimental results:

- Individuals prefer L_A to L'_A even if the expected monetary outcome from L'_A is larger (*certainty effect / loss aversion*).
- They also prefer L'_B to L_B , i.e. they are more willing to gamble for a bigger prize (*FOMO*).

Expected Utility Theory prediction: given $u_1 = u(1M)$, $u_2 = u(5M)$, $u_3 = u(0M)$, we have

$$U(L_A) = u_1 > U(L'_A) = 0.89u_1 + 0.1u_2 + 0.01u_3$$

but adding $-0.89u_1 + 0.89u_3$ to both members, we find

$$\begin{aligned} 0.11u_1 + 0.89u_3 &> 0.1u_2 + 0.9u_3 \\ \implies U(L_B) &> U(L'_B). \end{aligned}$$

Violation of independence

This paradoxical situation is due to a *violation of the independence axiom*.

Let us define degenerate lotteries L_i on outcomes with utility u_i and lotteries L and L' such that

$$U(L) = 1u_1, \quad U(L') = \frac{10}{11}u_2 + \frac{1}{11}u_3$$

Then we have

$$L_A \sim 0.11L + 0.89L_1, \quad L'_A \sim 0.11L' + 0.89L_1$$

and

$$L_B \sim 0.11L + 0.89L_3, \quad L'_B \sim 0.11L' + 0.89L_3$$

For the independence axiom: if $L \succsim L'$ (resp. $L' \succsim L$) then $L_A \succsim L'_A$ and $L_B \succsim L'_B$ (resp. $L'_A \succsim L_A$ and $L'_B \succsim L_B$).

Decisions under ignorance

The probability of the stochastic events is **unknown**.

Because of **ignorance**, there is no optimal criterion.

MAXIMIN CRITERION (WALD): the decision maker compares the minimum utility values provided by each alternative and chooses the one that maximizes the minimum value.

MINIMAX REGRET CRITERION (SAVAGE): the regret of an outcome is the difference between the maximum utility obtainable for some probability outcome and the utility value for that particular option; then the decision maker chooses the alternative for which the maximum regret is minimal.

PRINCIPLE OF INSUFFICIENT REASONS (LAPLACE): in the absence of prior information, all outcomes are assumed to be equally probable.

Decisions under ignorance

Example

Two friends are cooking an omelet. The first friend has just broken 5 good eggs into a bowl, then the second one is left to finish the job with the last egg. The latter does not know whether the sixth egg is good or rotten (probability p is not known).

The utilities of the outcomes $o_{xj} \in O$ are:

	good (prob. p)	rotten (prob. $1 - p$)
break into bowl	6-egg omelet (120)	throw everything away (0)
break into saucer	6-egg om. + saucer to wash (110)	5-egg om. + saucer to wash (90)
throw away	5-egg om. + good egg wasted (80)	5-egg omelet (100)

Which action is “best” according to the different criteria?